

Hybrid Product Recommendation System using Collaborative Filtering, FP-Growth and Sequential User Behavior Analysis

Phase 1: Problem Definition

Research Problem

Traditional recommendation systems often rely only on ratings or purchase history, which limits recommendation quality.

This project aims to improve recommendation accuracy by combining multiple user behaviors including:

- Ratings
- Purchase history
- Search history
- Browsing history
- Recently viewed products
- Shopping carts
- Wish lists
- Feedback
- Product popularity

Objective

Develop a hybrid recommendation engine that provides personalized product suggestions using multiple data mining techniques.

Phase 2: Dataset Collection

Required Data

User Information

- User ID
- Age
- Gender
- Preferences

Product Information

- Product ID
- Category
- Brand
- Price
- Description

Interaction Data

- Ratings
- Reviews
- Purchases
- Views
- Clicks
- Searches
- Wishlist items
- Cart items

Phase 3: Database Design

Tables

Users

Products

Ratings

Purchases

Cart

Wishlist

Search_History

Browsing_History

Reviews

Recently_Viewed

Transactions

Phase 4: Data Preprocessing

Tasks

Missing value handling

Duplicate removal

Outlier detection

Encoding categorical values

Normalization

Feature engineering

Session creation

User-item matrix generation

Phase 5: Collaborative Filtering

Purpose

Recommend products based on similar users.

Methods

User-Based Collaborative Filtering

Item-Based Collaborative Filtering

KNN

Cosine Similarity

Output

People similar to you purchased:

Product A

Product B

Product C

Phase 6: Matrix Factorization

Algorithm

Singular Value Decomposition (SVD)

Purpose

Discover hidden user preferences.

Examples

Luxury preference

Gaming interest

Budget sensitivity

Brand loyalty

Output

Predicted ratings

Top-N recommendations

Phase 7: Association Rule Mining

Algorithm

FP-Growth

Goal

Find products frequently purchased together.

Examples

iPhone

→ Cover

→ Charger

Laptop

→ Mouse

→ Keyboard

Metrics

Support

Confidence

Lift

Output

Frequently Bought Together

Phase 8: Sequential Pattern Mining

Algorithm

PrefixSpan

Purpose

Analyze order of user actions.

Example

Search

↓

View

↓

Wishlist

↓

Cart

↓

Purchase

Output

Next likely product recommendation

Behavior prediction

Phase 9: Search Behavior Analysis

Analyze

Search keywords

Frequency

Recent searches

Category interests

Output

Personalized search-based recommendations

Phase 10: Browsing Analysis

Data

Page visits

Viewing duration

Repeated visits

Recently viewed products

Output

Interest-based recommendations

Phase 11: Wishlist Analysis

Identify

Favorite categories

Preferred brands

Long-term interests

Output

Wishlist-driven recommendations

Phase 12: Shopping Cart Analysis

Detect

Products added together

Cart abandonment

Cross-selling opportunities

Output

Cart recommendations

Bundle suggestions

Phase 13: Popularity Analysis

Metrics

Most viewed

Most purchased

Highest rated

Trending products

Output

Trending section

Popular recommendations

Phase 14: Hybrid Recommendation Engine

Combine

Collaborative Filtering

-

SVD

-

FP-Growth

-

PrefixSpan

-

Cosine Similarity

Recommendation Score

Final Score

=

CF Score

•

Association Score

•

Sequential Score

•

Popularity Score

•

Similarity Score

Rank products

Generate Top-N recommendations

Phase 15: Evaluation

Metrics

Precision@K

Recall@K

F1-score

MAP

NDCG

Hit Rate

RMSE

Coverage

Diversity

Novelty

Phase 16: Frontend

Features

Home Page

Login

Search

Product Page

Recommendations

Wishlist

Cart

Trending Section

User Dashboard

Phase 17: Backend

Technology

Python

Flask / Django

MySQL

PostgreSQL

Pandas

Scikit-learn

Surprise Library

mlxtend

Phase 18: Final Outputs

Personalized Recommendations

Similar Products

Frequently Bought Together

Trending Products

Cart Recommendations

Wishlist Recommendations

Search-Based Recommendations

Top-N Product Suggestions

Expected Outcome

A complete hybrid recommendation system capable of analyzing user behavior, purchasing history, search activities, ratings, and browsing patterns to provide accurate personalized product recommendations similar to modern e-commerce platforms.

Algorithm Type Classification

Algorithm	Type	Category
Collaborative Filtering	Recommendation Algorithm	Recommender System
User-Based CF	Memory-Based	Collaborative Filtering
Item-Based CF	Memory-Based	Collaborative Filtering

Cosine Similarity	Similarity Measure	Distance/Similarity
KNN	Machine Learning	Neighbor-Based Recommendation
SVD	Matrix Factorization	Model-Based Collaborative Filtering
FP-Growth	Association Rule Mining	Data Mining
Apriori (Alternative)	Association Rule Mining	Data Mining
PrefixSpan	Sequential Pattern Mining	Sequence Mining
SPADE	Sequential Pattern Mining	Sequence Mining
Popularity Ranking	Ranking Algorithm	Recommendation
Top-N Ranking	Ranking Algorithm	Recommendation

The proposed system follows a Hybrid Recommendation Approach by integrating four major algorithmic categories:

- 1. Collaborative Filtering**
- 2. Association Rule Mining**
- 3. Sequential Pattern Mining**
- 4. Similarity-Based Ranking**

Collaborative Filtering is employed to identify users or items with similar preferences.

FP-Growth is utilized to discover frequently co-purchased products.

Sequential Pattern Mining (PrefixSpan) is used to analyze user behavioral sequences.

Cosine Similarity and ranking mechanisms are applied to generate Top-N personalized recommendations.

Hybrid Recommendation Algorithm

- **Collaborative Filtering**
- **Association Rule Mining**
- **Sequential Pattern Mining**
- **Similarity-Based Ranking**
- **Model-Based Recommendation (SVD)**